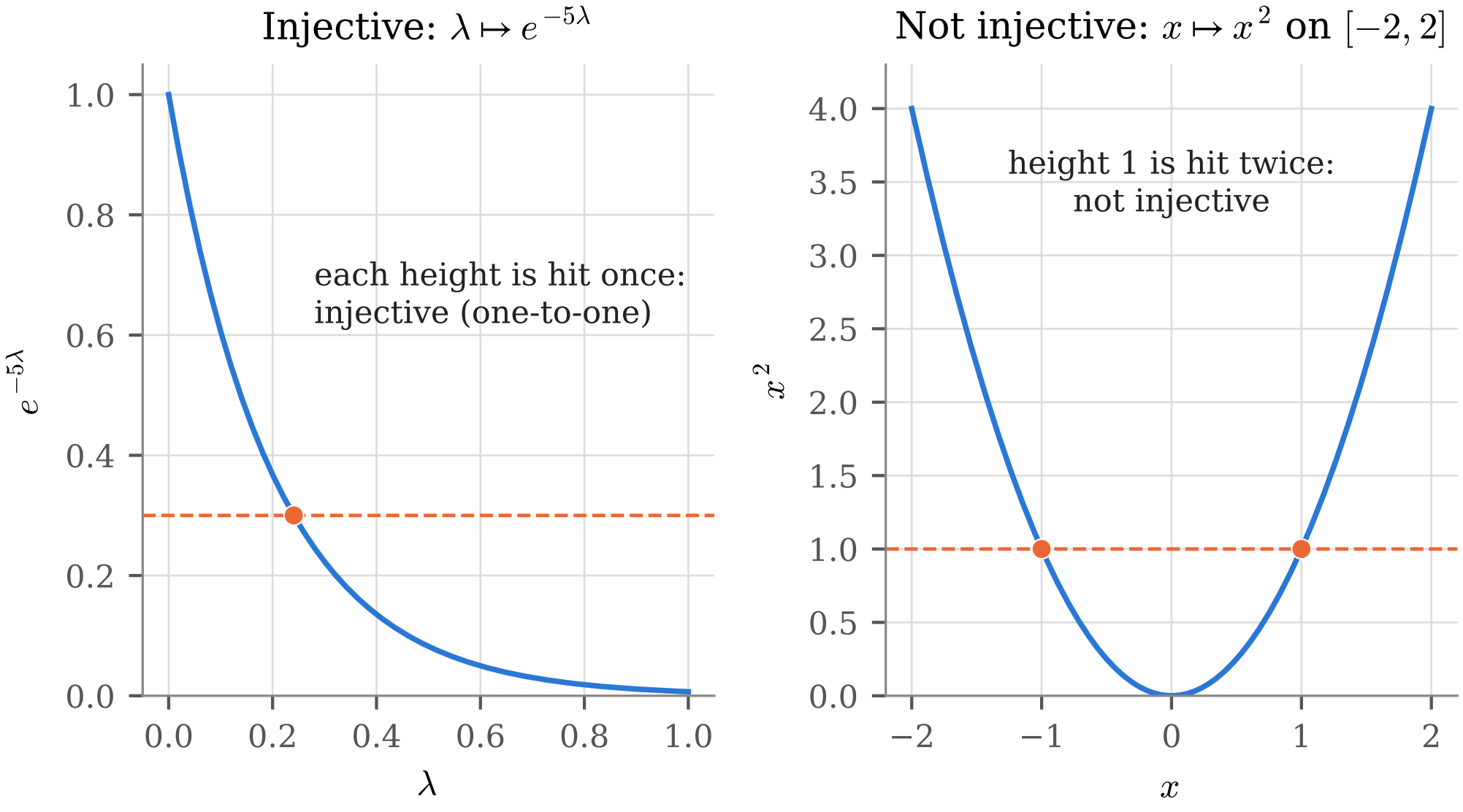
Injective: $\lambda \mapsto e^{-5\lambda}$
each height is hit once:
injective (one-to-one)
$e^{-5\lambda}$
λ
0.0
0.2
0.4
0.6
0.8
1.0
Not injective: $x \mapsto x^2$ on $[-2, 2]$
height 1 is hit twice:
not injective
x^2
x
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
−2
−1
0
1
2